

SALEN-REUTENEN, Switzerland

WMO: 066230

Lat: 47.6511N

Lon: 9.0239E

Elev: 2361

StdP: 13.48

Time Zone: 1.00 (EUC)

Period: 11-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	13.1	18.7	13.6	12.0	21.1	16.5	13.8	22.6	29.9	43.8	26.5	38.9	8.0	60	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.5	83.2	66.3	80.1	64.8	77.0	63.8	68.4	79.1	66.8	77.3	65.3	74.9	5.9	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.0	101.0	73.4	63.1	94.3	70.8	61.8	89.9	69.2	34.1	79.5	32.8	77.5	31.5	75.3	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.8	20.5	17.3	DB	12.1	87.6	6.9	2.2	7.1	89.2	3.1	90.4	-0.8	91.7	-5.8	93.2	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.7	32.6	32.3	41.0	48.2	54.1	62.1	65.2	64.7	57.4	49.5	40.2	35.3
	DBStd	13.69	7.04	9.00	7.26	8.15	7.15	6.95	6.61	6.47	5.87	7.05	6.97	6.26
	HDD50	2357	539	495	288	131	41	1	0	0	5	97	305	455
	HDD65	6198	1004	915	744	505	344	136	80	88	238	479	745	920
	CDD50	1865	0	0	9	77	168	365	471	456	226	83	10	0
	CDD65	231	0	0	0	1	6	50	86	79	9	0	0	0
	CDH74	1397	0	0	0	5	25	343	538	466	20	0	0	0
	CDH80	277	0	0	0	0	0	67	121	89	0	0	0	0
Wind	WSAvg	7.1	8.9	8.1	8.3	6.6	7.0	6.1	6.3	5.6	6.1	6.4	7.6	8.6
Precipitation	PrecAvg	38.2	2.2	2.1	2.4	2.7	4.2	4.5	4.4	4.0	3.3	2.8	2.8	2.8
	PrecMax	47.8	5.8	4.8	7.5	5.5	8.4	7.4	8.5	7.4	6.2	5.3	5.7	6.4
	PrecMin	29.1	0.2	0.6	0.7	0.4	0.8	1.3	1.7	1.2	1.4	0.4	0.0	0.1
	PrecStd	4.6	1.5	1.2	1.4	1.3	1.9	1.6	1.7	1.6	1.3	1.3	1.5	1.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.3	54.7	62.5	72.9	77.3	85.7	87.7	86.1	76.7	67.3	60.7	53.7
		MCWB	45.0	43.8	48.9	56.2	61.5	67.6	67.8	66.8	62.1	56.3	50.4	43.9
	2%	DB	47.4	49.6	59.2	68.2	72.7	81.9	83.2	82.5	73.4	64.2	56.8	49.3
		MCWB	43.3	42.6	47.1	52.9	59.4	65.5	65.7	66.7	62.0	55.9	50.0	44.8
	5%	DB	44.8	46.5	55.4	65.4	69.3	77.9	79.9	79.4	70.1	61.7	53.8	46.9
		MCWB	41.2	41.1	45.5	52.3	57.5	64.1	64.0	65.7	61.0	55.2	49.2	43.9
	10%	DB	41.6	43.7	52.1	61.5	66.1	74.3	76.2	75.9	67.0	59.1	50.5	44.7
		MCWB	39.8	39.4	43.9	50.5	55.5	62.9	63.1	64.5	59.5	54.3	47.2	42.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.2	47.5	50.7	57.7	64.5	70.6	71.1	70.8	64.6	59.6	54.2	48.0
		MCDB	48.8	50.1	59.7	69.7	72.7	81.9	82.8	79.3	72.9	63.9	58.5	49.2
	2%	WB	44.3	44.1	48.3	54.9	61.8	67.6	68.0	68.4	63.0	57.9	51.3	45.9
		MCDB	46.0	48.9	56.5	66.3	69.6	78.5	80.0	78.2	70.7	62.1	54.8	47.6
	5%	WB	41.9	41.9	46.0	52.9	59.4	65.8	66.4	66.7	61.7	56.3	49.4	44.3
		MCDB	43.9	46.4	52.4	63.1	65.9	75.8	76.7	76.8	68.3	60.1	52.7	47.2
	10%	WB	40.1	39.9	44.5	51.4	57.1	63.9	64.7	65.2	60.2	54.9	47.6	42.6
		MCDB	41.9	43.5	49.9	60.6	63.8	72.6	74.0	74.7	65.7	58.2	50.6	45.2
Mean Daily Temperature Range	5% DB	MDBR	6.3	8.5	12.7	14.3	14.3	14.6	14.5	14.4	12.4	9.8	7.3	6.7
		MCDBR	9.7	13.9	19.1	19.5	19.0	19.0	18.6	18.8	15.7	12.7	11.7	8.8
		MCWBR	7.4	8.5	9.1	8.9	8.9	9.3	7.4	7.7	7.3	7.7	7.2	5.9
	5% WB	MCDBR	9.0	12.6	14.6	16.9	16.3	18.2	16.7	16.8	14.1	11.6	10.6	7.8
		MCWBR	7.3	8.6	8.6	8.3	8.8	9.5	7.7	8.3	7.0	7.8	7.6	6.3
Clear-Sky Solar Irradiance	taub	0.282	0.304	0.336	0.369	0.381	0.386	0.388	0.382	0.358	0.332	0.303	0.277	
	taud	2.467	2.408	2.386	2.323	2.324	2.342	2.342	2.328	2.363	2.413	2.477	2.508	2.471
	Ebn at Noon	258	273	280	280	280	278	276	273	269	260	247	245	
	Edh at Noon	20	27	32	37	39	38	38	36	31	25	20	18	
All-Sky Solar Radiation	RadAvg	352	613	996	1367	1586	1817	1746	1525	1139	711	391	288	
	RadStd	39	86	134	195	161	166	159	157	118	76	47	45	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (112 neighbors)	N/A	N/A	N/A	+1.24	+0.77	+0.65	N/A	N/A	+99	+25

Nomenclature: See separate page